www.tntextbooks.in

of positron emission from radiolabelled tracer molecules. These radiotracers allow biologic processes to be measured and whole body images to be obtained which demonstrates sites of radiotracer accumulation. A PET image gives quantitative regional information on the metabolic and physiological processes. PET uses positron emitting radio isotopes (^{11}C ^{13}N ^{15}O ^{18}F) which are generated by the cyclotron. The most common radiotracer in use today is 18F-fluorodeoxyglucose (18F-FDG) which is a radio labelled sugar (glucose) molecule. These atoms are then incorporated by chemical methods into biological molecules like glucose, amino acids and ammonia. These positron emitting compounds are then injected in very small amounts into or inhaled by experimental animals or human subjects. The three dimensional distribution of the labeled trace is then probed by powerful PET cameras and the images are reconstructed by a computer. The quantitative interpretation of the image is done by varying mathematical models. They deal with the process of uptake and metabolism of the radioisotope.

Clinical significance

PET imaging is effectively used in the measurement of regional cerebral blood volume, blood flow, metabolic rates for glucose and oxygen in humans.

NMRI-Magnetic Resonance Imaging

Magnetic resonance imaging (MRI) is a non-invasive medical test that physicians use to diagnose medical conditions. Unlike conventional X-ray examinations and computed tomography (CT) scans, MRI does not utilize ionizing radiation. MRI uses a powerful magnetic field, radio frequency pulses and a computer to produce detailed pictures of organs, soft tissues, bone and virtually all other internal body structures. The radio frequency pulses re-align hydrogen atoms that naturally exist within the body while the patient is in the scanner without causing any chemical changes in the tissues. As the hydrogen atoms return to their usual alignment, they emit different amounts of energy that vary according to the type of body tissue from which they come. The MR scanner captures this energy and creates a picture of the tissues scanned based on this information (Figure12.9).

The magnetic field is produced by passing an electric current through wire coils in most MRI units. Other coils, located in the machine and in some cases, are placed around the part of the body being imaged, send and receive radio waves, producing signals that are detected by the coils. The electric current does not come in contact with the patient.

A computer then processes the signals and generates a series of images, each of which shows a thin slice of the body. The images can then be studied from different angles and interpreted by a radiologist.

Frequently, the differentiation of abnormal (diseased) tissue from normal tissues is better with MRI than with other imaging modalities such as X-ray, CT and ultrasound. Detailed MR images allow physicians to evaluate various parts of the body and determine the presence of certain diseases. The images can then be examined on a computer monitor, transmitted electronically, printed or copied to a CD or uploaded to a digital cloud server.